Annexure-I

Associate Software Engineer (NOC Intern)

MAQ Software

A training report

Submitted in partial fulfillment of the requirements for the award of degree of

B.tech

Computer Science and Engineering

Submitted to

LOVELY PROFESSIONAL UNIVERSITY

PHAGWARA, PUNJAB **From 07/17/25 to 12/31/25**

SUBMITTED BY

Name of student: Shreya Choudhary

Registration Number: 12208155

Annexure-II: Student Declaration

To whom so ever it may concern

I, **Shreya Choudhary, 12208155,** hereby declare that the work done by me on "**Associate Software Engineer (NOC Intern)** " from **Nov, 2025** to **Ongoing**, under the supervision of **Sir Nandakishore Mohan**, **Assistant Technical Project Manager, MAQ Software** , is a record of original work for the partial fulfillment of the requirements for the award of the degree, **B.tech (CSE).**

Name of the Student (Registration Number): **Shreya Choudhary (12208155)**

Dated: **31/12/2025**

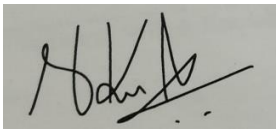
Annexure-III: Declaration by the supervisor/Mentor

To whom so ever it may concern

This is to certify that **Shreya Choudhary**, **12208155** from Lovely Professional University, Phagwara, Punjab, has worked as a trainee in **MAQ Software** on "**Associate Software Engineer (NOC Intern)**" under my supervision from **Nov, 2025** to **Ongoing**. It is further stated that the work carried out by the student is a record of original work to the best of my knowledge for the partial fulfillment of the requirements for the award of the degree, **B.tech(CSE)**.

Name of External Supervisor: **Nandakishore Mohan**

Designation of the External Supervisor: **Assistant Technical Project Manager**



Supervisor Signature of the external Supervisor

Dated: **31/12/2025**

Annexure IV

CONTINUOUS ASSESSMENT (CA) for INTERNSHIP/OJT

(By external Supervisor from organization)

CONTINUOUS ASSESSMENT (CA) for INTERNSHIP/OJT

(By external Supervisor from organization)

Name of the student **Shreya Choudhary**, Registration Number **12208155**
Internship Project Title (if/any): **GPS Data Ops**
Name of Organization &Address: MAQ Software, A3, Sector 145, Near Metro Station, Noida, G. B. Nagar, UP 201301

Name of External Internship in-charge (with mobile number):
Nandakishore Mohan Contact No: +91 9961248729

S.No.	Criteria	Marks Obtained	Maximum Marks
1	Student conduct during internship	9	10
2	Punctuality and Enthusiasm	17	20
3	Technical Skill & Knowledge	18	20
4	Performance	46	50
	Total	**90**	**100**

Date 25/03/2026

Authorized Signatory

Name: Nandakishore Mohan
Designation Associate Technical Project Manager

MAQ INDIA Private Limited Noida

Company Seal

Annexure IV

Salary Slip

MAQ Software

MAQ India Private Limited
201, Meadows Building, Sahar Plaza, Andheri Kurla Road Mumbai Maharashtra 400059 India

Payslip For the Month
November 2025

EMPLOYEE SUMMARY

Employee Name	:	Shreya Choudhary
Designation	:	Associate Software Engineer
Employee ID	:	EMPN2130
Date of Joining	:	November 03, 2025
Pay Period	:	November 2025
Pay Date	:	November 28, 2025

₹19,974.00
Total Net Pay

Paid Days	:	28
LOP Days	:	0

Bank Account No	:	50100841285810	PAN	:	CTHPC6335A
Work Location	:	Noida			
Aadhar Number	:	444081414319			

EARNINGS	AMOUNT	YTD	DEDUCTIONS	AMOUNT	YTD
Basic	**₹14,000.00**	₹14,000.00	EPF Contribution	**₹1,680.00**	₹1,680.00
House Rent Allowance	**₹6,487.00**	₹6,487.00			
Bonus	**₹1,167.00**	₹1,167.00			
Gross Earnings	**₹21,654.00**		**Total Deductions**	**₹1,680.00**	

TOTAL NET PAYABLE
Gross Earnings - Total Deductions
₹19,974.00

Amount In Words : Indian Rupee Nineteen Thousand Nine Hundred Seventy-Four Only

-- This is a system-generated document. --

End-Term Practical Training Report

Internship Details

- **Company Name:** MAQ Software
- **Role:** Associate Software Engineer
- **Department:** DataOps
- **Industry:** MS Azure & Fabric
- **Duration:** One Year Internship
- **Location:** Noida, UP

1. Introduction

I had the incredible opportunity to intern at MAQ Software, a global technology consulting firm that focuses on delivering data analytics and cloud-based solutions. The company works extensively with platforms like Microsoft Azure and Microsoft Fabric to help organizations make data-driven decisions and improve operational efficiency.

As a DataOps Engineer Intern, I worked on monitoring pipeline runs and managing data refresh processes, ensuring that data was processed correctly and made available for downstream users. After refreshes, we processed the data so it could be consumed in the form of reports, cubes, and structured views within the data mart in a lakehouse architecture. I also handled ad hoc tasks and supported data engineering workflows, primarily working with SQL and SQL Server Management Studio.

This experience allowed me to understand how data pipelines function in real-world scenarios and how processed data is used for business insights. It also helped me develop practical skills in SQL, problem-solving, and working within a production environment, bridging the gap between academic learning and industry practices.

2. Internship Objectives

When I started my internship at MAQ Software, I had several clear objectives in mind that I wanted to achieve:

- **Understanding Data Systems:** I aimed to comprehend the end-to-end flow of data pipelines within cloud platforms like Microsoft Azure and Microsoft Fabric, and how data moves from ingestion to final consumption in real-world scenarios.

● **Mastering DataOps Processes:** I wanted to learn how to monitor pipeline runs, manage data refresh cycles, and handle failures effectively in a production environment.

● **Developing Collaboration Skills:** I sought to work closely with data engineers and stakeholders to ensure smooth execution of workflows and timely resolution of issues.

● **Building Incident Handling Expertise:** I aimed to develop the ability to identify, troubleshoot, and resolve data-related issues efficiently, especially during critical pipeline failures or refresh delays.

● **Enhancing Technical Competencies:** I focused on strengthening my skills in SQL and tools like SQL Server Management Studio, along with gaining hands-on experience in data modelling and working with Microsoft Power BI reports and datasets.

● **Gaining Industry Exposure:** I wanted to understand how large-scale data systems are maintained and how processed data is delivered to downstream users through reports, cubes, and structured views in a lakehouse architecture.

Throughout my internship, I actively worked toward these objectives and was able to make meaningful progress across each of these areas.

3. Overview of GreyOrange's Automation Framework

Understanding the data and analytics framework at MAQ Software was a key part of my internship. The company focuses on accelerating business intelligence and analytics initiatives for enterprises using cloud and AI-driven solutions, primarily built on Microsoft Azure and modern analytics platforms like Microsoft Fabric. Their ecosystem is designed to transform raw data into actionable insights that help organizations improve productivity, reduce costs, and make better business decisions.

3.1 Core Components

● **Cloud & Data Platforms:** MAQ Software leverages Azure for scalable cloud infrastructure and Fabric as an integrated analytics platform that combines data engineering, data integration, and real-time analytics in one unified system.

● **Business Intelligence (Power BI):** A major component of their framework is Microsoft Power BI, which is used to create interactive dashboards, reports, and data models for enterprise-level decision-making.

● **Data Engineering & Lakehouse Architecture:** The system centralizes data into unified storage (lakehouse), where it is cleaned, transformed, and structured for analytics and reporting across the organization.

● **AI & Analytics Solutions:** The company integrates AI and advanced analytics into its solutions to enable predictive insights and smarter decision-making across business processes.

3.2 System Integration

What stood out was how seamlessly these components are integrated. Data flows through cloud-based pipelines into centralized storage systems, where it is processed and modeled before being visualized through Power BI dashboards and reports. This unified approach ensures that data remains consistent, accessible, and ready for consumption across different teams and business functions.

3.3 Operational Criticality

Since MAQ Software delivers business-critical analytics solutions to enterprises, maintaining the accuracy and availability of data systems is essential. Any disruption in data pipelines or reporting layers can directly affect business insights and decision-making. This highlights the importance of continuous monitoring, efficient issue resolution, and maintaining high-performance data systems within such an ecosystem.

4. Role & Key Responsibilities

4.1 Position Overview: DataOps Engineer Intern

My role as a DataOps Engineer Intern at MAQ Software placed me at the core of data operations and analytics workflows. I was responsible for monitoring and maintaining data pipelines, ensuring smooth data refresh cycles, and supporting the availability of data for business intelligence systems. This role required close coordination with data engineering teams and a strong focus on ensuring that data was accurate, up-to-date, and ready for downstream consumption.

4.2 Daily Responsibilities

My typical workday involved a range of operational and analytical tasks:

● **Pipeline Monitoring:** I continuously monitored pipeline runs within Microsoft Azure and Microsoft Fabric to ensure successful execution and identify any failures or delays.

● **Data Refresh Management:** I tracked and validated data refresh cycles to ensure that updated data was correctly processed and available for reporting.

● **Issue Identification & Troubleshooting:** I performed initial analysis of failed pipelines or refresh issues, identifying root causes and assisting in resolving them efficiently.

● **Ad Hoc Task Handling:** I worked on dynamic requests related to data validation, query execution, and quick fixes based on business requirements.

● **SQL Operations:** I extensively used SQL Server Management Studio (SSMS) to run queries, analyze datasets, and support data engineering tasks.

● **Data Processing Support:** After successful refreshes, I ensured that data was processed and structured correctly for downstream usage in reports, cubes, and views.

● **Power BI & Data Modelling:** I supported work related to Microsoft Power BI reports and data models, helping ensure accurate and efficient visualization of data.

4.3 Tools and Systems Used

During my internship, I gained hands-on experience with modern data platforms and tools used in enterprise analytics environments. I worked extensively with Azure and Fabric for pipeline monitoring and data workflows, SSMS for SQL-based operations, and Power BI for reporting and data modelling. These tools provided real-time insights into data systems and helped maintain the smooth functioning of end-to-end data processes.

5. Ticket Triage & Severity Framework

5.1 Understanding the Workflow

One of the key aspects of my role as a DataOps Engineer Intern at MAQ Software was understanding how to monitor and manage data-related issues in production systems. Issues typically originated from disruptions in data workflows such as:

● **Pipeline Failures:** Scheduled or triggered pipelines in Microsoft Azure or Microsoft Fabric failing due to configuration errors, dependency issues, or runtime exceptions.

● **Data Refresh Failures:** Incomplete or failed refreshes affecting downstream datasets and reports.

● **Data Quality Issues:** Missing, inconsistent, or incorrect data appearing in processed outputs.

● **SQL & Data Processing Errors:** Query failures or incorrect transformations while working in SQL Server Management Studio.

● **Reporting Issues:** Discrepancies or delays in Microsoft Power BI reports and dashboards due to upstream data problems.

My approach involved analyzing each issue based on:

● **Nature of the Issue:** Understanding whether the problem originated from pipelines, data processing, or reporting layers.

● **Impact on Downstream Systems:** Evaluating how the issue affected reports, cubes, and data views used by stakeholders.

● **Urgency of Resolution:** Determining how quickly the issue needed to be fixed based on business dependency on the data.

● **Dependency Analysis:** Identifying upstream and downstream dependencies to trace root causes effectively.

● **Deep Data Analysis:** In cases such as BVT (Build Verification Test) failures or row count mismatches, I performed detailed data analysis using SQL to compare datasets, validate transformations, and identify where discrepancies were introduced in the pipeline.

5.2 Severity Classification System

To manage issues efficiently, I learned to classify them into different severity levels:

● **Critical (Severity 1):** Complete failure of pipelines or data refreshes impacting major reports or business-critical dashboards, requiring immediate attention.

● **High (Severity 2):** Significant data delays or partial failures affecting key datasets or reports, impacting business decisions but not halting everything.

● **Medium (Severity 3):** Issues like intermittent failures, minor data inconsistencies, row count mismatches, or performance slowdowns with manageable impact.

● **Low (Severity 4):** Minor issues such as small data discrepancies, non-critical report issues, or ad hoc requests that do not significantly impact business operations.

5.3 Importance of Correct Classification

Correctly identifying the severity of issues was crucial. Overestimating severity could lead to unnecessary escalations, while underestimating could impact business reporting and decision-making. This process helped me develop strong analytical thinking and decision-making skills while working in a real-time data operations environment.

6. Ticket Assignment & Team Collaboration

6.1 Specialized Teams

After identifying and analyzing issues, tasks needed to be aligned with the appropriate teams based on their area of expertise. Understanding team responsibilities was important for efficient resolution:

● **Data Engineering Teams:** Handled pipeline failures, transformation logic issues, and overall data workflow problems within platforms like Microsoft Azure and Microsoft Fabric.

● **BI & Reporting Teams:** Managed issues related to dashboards, reports, and data models in Microsoft Power BI.

● **Database Teams:** Addressed SQL-related issues, performance bottlenecks, and data inconsistencies using tools like SQL Server Management Studio.

● **Platform & Infrastructure Teams:** Took care of environment-level issues, access problems, and system-level configurations.

6.2 Understanding System Architecture

To ensure correct handling of issues, I developed a strong understanding of how data systems were structured—from pipeline execution to data processing and final reporting layers. This helped me identify where a failure originated and

which team should handle it. Avoiding misalignment was important, as incorrect routing could delay resolution and impact downstream users relying on the data.

6.3 Collaboration Experiences

Working with multiple teams helped me build effective communication and collaboration skills. I learned to:

● Provide clear and concise context while reporting or escalating issues
● Share relevant data points, logs, and SQL findings to support debugging
● Coordinate across teams when issues spanned multiple layers (pipeline, database, reporting)
● Follow up on critical issues while being mindful of team priorities
● Build smooth working relationships that improved overall efficiency and turnaround time

7. Incident Management

7.1 Introduction to Incident Management

One of the most valuable aspects of my internship at MAQ Software was gaining exposure to incident management in data operations—a structured approach to handling critical failures in pipelines, data refreshes, and reporting systems that require immediate attention and coordination across teams.

7.2 When Incidents Are Declared

Not every issue is treated as an incident. Incidents are typically declared when:

● Multiple critical pipelines or refresh processes fail simultaneously
● Business-critical reports or dashboards in Microsoft Power BI become unavailable or show incorrect data
● There is a major impact on downstream users relying on data for decision-making
● Issues persist despite initial troubleshooting and require escalation
● Upstream data changes impact existing pipelines, requiring an ICM (Incident Communication Management) to be raised and coordinated with stakeholders at Microsoft

7.3 Coordinated Response Approach

For critical issues, a coordinated response involving multiple teams is required. This includes data engineers, BI teams, and platform teams working together to identify root causes and restore normal operations.

This approach focuses on:

● **Real-time Collaboration:** Teams work together to analyze failures in Microsoft Azure and Microsoft Fabric pipelines and resolve them quickly

● **Transparent Communication:** Regular updates are shared among stakeholders, including external coordination when upstream dependencies are involved

● **Rapid Decision-Making:** Teams align quickly on fixes, whether it involves rerunning pipelines, adjusting transformations, or accommodating upstream changes

● **Coordinated Execution:** Ensuring all teams work in sync across ingestion, processing, and reporting layers

7.4 My Participation

Initially, I observed how incidents were handled to understand workflows and communication patterns. Over time, I started contributing by monitoring impacted pipelines, sharing SQL-based analysis, identifying BVT or row count mismatches, and helping track resolution progress. I also assisted in raising ICMs for upstream data changes and communicating with stakeholders to ensure alignment and timely resolution. This experience strengthened my ability to work in high-impact scenarios within real-time data systems.

9. Technical & Professional Skills Acquired

9.1 Technical Competencies Developed

● **Data Platforms & Tools:** Gained hands-on experience working with Microsoft Azure and Microsoft Fabric, understanding how enterprise-level data systems are built and maintained.

● **SQL & Data Analysis:** Strengthened my proficiency in SQL using SQL Server Management Studio, including querying, debugging, and performing deep data analysis for issues like BVT failures and row count mismatches.

● **Power BI & Data Modelling:** Worked with Microsoft Power BI reports and datasets, gaining exposure to data modelling and how structured data is used for business insights.

● **Pipeline Monitoring & Troubleshooting:** Developed the ability to monitor pipeline runs, identify failures, and troubleshoot issues across different stages of data workflows.

● **Data Validation & Quality Checks:** Learned to validate data consistency and accuracy by comparing datasets, verifying transformations, and ensuring reliable outputs for downstream consumption.

● **Understanding Data Architecture:** Built a strong understanding of lakehouse architecture, data marts, and how data flows from ingestion to reporting layers.

9.2 Soft Skills Enhanced

● **Professional Communication:** Improved my ability to communicate effectively with both technical teams and stakeholders, especially while explaining issues and updates clearly.

● **Cross-team Collaboration:** Developed the ability to work with data engineers, BI teams, and external stakeholders, ensuring smooth coordination during issue resolution.

● **Time & Priority Management:** Learned to manage multiple tasks simultaneously, prioritizing critical pipeline failures and refresh issues over less urgent work.

● **Problem-Solving Mindset:** Strengthened my analytical thinking by approaching issues methodically and identifying root causes efficiently.

● **Adaptability:** Became comfortable handling dynamic tasks, ad hoc requests, and evolving priorities in a production environment.

● **Attention to Detail:** Built a strong focus on accuracy while working with data, ensuring that even minor discrepancies were identified and resolved.

● **Working Under Pressure:** Learned to stay composed and effective during critical failures and high-impact scenarios, maintaining consistency in performance.

10. Challenges Faced and How I Overcame Them

10.1 Initial Learning Curve

Coming from a full-stack development background, transitioning into a DataOps role at MAQ Software was initially challenging. I had to shift my mindset from building applications to understanding data pipelines, cloud platforms like Microsoft Azure and Microsoft Fabric, and large-scale data workflows. I overcame this by:

● Actively learning how data systems function end-to-end
● Revisiting SQL concepts and applying them practically using SQL Server Management Studio
● Studying existing pipelines and understanding their dependencies
● Gradually adapting to thinking in terms of data flow rather than application logic

10.2 Handling Pressure

Working in a production data environment meant handling time-sensitive issues like pipeline failures and delayed data refreshes that directly impacted reports and stakeholders. Initially, this pressure was new to me, but I managed it by:

● Following a structured approach to troubleshooting instead of reacting impulsively
● Prioritizing issues based on their impact on downstream systems
● Collaborating with team members during critical situations
● Staying calm and focused while resolving high-priority issues

10.3 Technical Knowledge Gaps

Since my prior experience was more aligned with full-stack development, I initially faced gaps in data engineering concepts, data modelling, and analytics workflows. I addressed this by:

● Being open about my learning curve and asking questions
● Learning SQL more deeply and applying it to real-world data problems

● Understanding concepts like lakehouse architecture and pipeline orchestration
● Building a habit of documenting learnings from each issue I encountered

10.4 Communication Challenges

Communicating data-related issues to stakeholders, especially when explaining pipeline failures or data discrepancies, was initially difficult. I improved this by:

● Simplifying technical explanations into clear, understandable updates
● Structuring my communication around impact and resolution rather than raw technical details
● Learning from feedback provided by team members
● Keeping communication concise, relevant, and focused on outcomes

11. Key Learnings & Outcomes

11.1 Technical Insights

● Gained a strong understanding of how large-scale data systems operate in real-world environments using platforms like Microsoft Azure and Microsoft Fabric
● Learned that maintaining reliable data pipelines and refresh cycles is as important as building them
● Understood the importance of monitoring, validation checks (like BVT and row counts), and proactive issue detection
● Developed clarity on how data flows from ingestion to final consumption in reports and analytics systems such as Microsoft Power BI

11.2 Professional Growth

● Developed a professional work ethic and understanding of working in a data-driven corporate environment at MAQ Software
● Learned the importance of documentation, structured workflows, and process adherence
● Built confidence in handling real-time production issues and contributing to business-critical systems
● Understood how different teams collaborate across data engineering, BI, and platform layers

11.3 Career Clarity

● Discovered a strong interest in DataOps, data engineering, and analytics workflows
● Realized that I enjoy troubleshooting data issues, performing analysis, and ensuring system reliability
● Identified potential career paths in DataOps, data engineering, and cloud-based data platforms
● Understood how skills in SQL, data modelling, and cloud systems can be applied in industry

11.4 Personal Development

- Became more confident in my technical decision-making and communication skills
- Developed the ability to adapt quickly to new tools, workflows, and problem scenarios
- Improved my ability to work under pressure during critical pipeline failures and incidents
- Gained appreciation for teamwork, collaboration, and effective stakeholder communication

12. Memorable Experiences

12.1 Handling a Critical Data Issue

One of the most memorable moments during my internship at MAQ Software was handling a critical data issue where key pipelines failed, impacting downstream reports and business visibility. The situation required quick analysis and coordination, and being part of the resolution process gave me valuable exposure to how high-impact data issues are managed in real time.

12.2 Successful Problem Solving

There was a particularly challenging case involving data discrepancies, where BVT checks and row counts were failing. By performing detailed SQL analysis using SQL Server Management Studio and tracing data across pipeline stages, I was able to identify the root cause. Contributing to resolving such an issue and receiving appreciation from the team was a very rewarding experience.

12.3 Team Collaboration

One of the best parts of my internship was the team I worked with. The collaboration across teams working on Microsoft Azure, Microsoft Fabric, and reporting systems was smooth and supportive. I had a great team that was always willing to help, share knowledge, and guide me whenever needed. Along with learning, I genuinely enjoyed working with them and had a really good overall experience during my internship.

13. Suggestions and Recommendations

Based on my experience, I would suggest:

- **For Future Interns:** Take initiative in learning, don't hesitate to ask questions, and embrace every challenge as a learning opportunity. The more you engage, the more you'll gain from the experience.

- **For the Organization:** Consider creating a more structured onboarding program for interns with graduated responsibility levels. Pair programming or "shadowing days" with different teams could provide broader exposure to the organization.

- **For Academic Institutions:** Strengthening practical, hands-on components in the curriculum would better prepare students for industry roles. Case studies based on real production issues would be particularly valuable.

15. Conclusion

My internship at MAQ Software as a DataOps Engineer Intern has been an intellectually enriching and professionally transformative experience that has significantly shaped my career direction. The exposure to real-time data operations, pipeline monitoring, incident handling, and cross-functional collaboration helped me understand how modern data systems operate at scale using platforms like Microsoft Azure and Microsoft Fabric.

Working in a DataOps environment taught me that building data systems is only one part of the process—ensuring their reliability, accuracy, and availability is equally important. I gained a deeper appreciation for monitoring, validation, and continuous improvement, especially in systems where downstream reports and business decisions depend heavily on data. The collaborative culture and structured workflows I experienced have set a strong benchmark for my future professional journey.

I am truly grateful to MAQ Software for providing me with this opportunity and for fostering an environment where I could actively contribute and learn. I would also like to thank my team and mentors for their constant support, guidance, and willingness to share knowledge throughout my internship.

I look forward to applying these learnings in the future and hope to contribute even more meaningfully in similar roles within the data and analytics domain.

Acknowledgments

I would like to express my sincere gratitude to:

- My manager for providing guidance and constructive feedback throughout my internship
- The entire team for their support and collaboration
- The dev teams who patiently explained complex concepts and shared their expertise
- MAQ's HR team for facilitating a smooth internship experience
- My academic institution for providing this practical training opportunity